(Task 1 of 10) In this tutorial, we will learn even more about definitions. 0

Lispy |

(Task 2 of 10) What is the result of running this program?

Lispy [Run ▶]

```
(defvar x (+ y 1))
(defvar y 2)

x
y
```

Scala 3

```
val x = y + 1
val y = 2
println(x)
println(y)
```

`3 2` 2

The answer is `error`. 3

▼ Textual explanation

You might think `(+ y 1)` is able to refer to `y` and hence evaluate to a value. However, `y` has not been bound to a value when `(+ y 1)` is evaluated. In SMoL, every block evaluates its definitions and expressions in reading order (i.e., top-to-bottom and left-to-right).

Click here to run this program in the Stacker.

► Graphical explanation

Lispy |

What is the result of running this program?

Lispy [Run ▶]

```
(defvar baz (+ bar 1))
(defvar bar 2)

baz
bar
```

JavaScript

```
let baz = bar + 1;
let bar = 2;
console.log(baz);
console.log(bar);
```

`error` 5

You predicted the output correctly 🎉🎉🎉 6

Lispy |

(Task 3 of 10) What is the result of running this program?

Lispy [Run ▶]

```
(deffun (f x)
  3)
(f (/ 7 0))
```

JavaScript

```
function f(x) {
  return 3;
}
```

```
console.log(f(7 / 0));
```

⚠️ JavaScript behaves differently.

`error` 8

You predicted the output correctly 🎉🎉🎉 9

Function calls bind their formal parameters (in this case, there is one formal parameter, `x`) to the values of actual parameters (in this case, there is one actual parameter, `(/ 7 0)`). The program errors when it tries to evaluate `(/ 7 0)`.

Click here to run this program in the Stacker.

Lispy | 🎲

(Task 4 of 10) What is the result of running this program?

Lispy [Run ▶️]	Python
`(defvar x (/ 12 0))` `3`	`x = 12 / 0` `print(3)`

⚠️ JavaScript behaves differently.

`3` 11

The answer is `error`. 12

▶ Textual explanation

Lispy | 🎲

What is the result of running this program?

Lispy [Run ▶️]	JavaScript
`(defvar b (/ 79 0))` `1`	`let b = 79 / 0;` `console.log(1);`

⚠️ JavaScript behaves differently.

`error` 14

You predicted the output correctly 🎉🎉🎉 15

16

(Task 5 of 10) In what order are definitions and expressions evaluated? 16

top to bottom 17

(Task 6 of 10) - Variables are bound to values. Specifically, every variable definition evaluates the expression immediately and binds the variable to the value, even if the variable is not used later in the program; every function call evaluates the actual parameters immediately and binds the values to formal parameters, even if the formal parameter is not used in the function. 18

- Every block evaluates its definitions and expressions in reading order (i.e., top-to-bottom and left-to-right).

Any feedback regarding these statements? Feel free to skip this question. 19

(You skipped the question.) 20

(Task 7 of 10) Please scroll back and select 1-3 programs that together make these points. 21

> - Variables are bound to values. Specifically, every variable definition evaluates the expression immediately and binds the variable to the value, even if the variable is not used later in the program; every function call evaluates the actual parameters immediately and binds the values to formal parameters, even if the formal parameter is not used in the function.

You don't need to select all such programs.

(*You selected 3 programs*) 22

Okay. How do these programs (4,7,13) support the point? 23

Show immediate eval 24

(Task 8 of 10) Please scroll back and select 1-3 programs that together make these points. 25

> - Every block evaluates in top-to-bottom, left-to-right order.

You don't need to select all such programs.

(*You selected 2 programs*) 26

Okay. How do these programs (1,4) support the point? 27

show top down - as cannot refer to lower var 28

Lispy |

(Task 9 of 10) Here is a program that confused many students 29

Lispy [Run ▶]

```
(deffun (addy x)
  (+ x y))
(defvar s (addy 1))
(defvar y 2)

s
```

Scala 3

```
def addy(x : Int) =
  x + y
val s = addy(1)
val y = 2
println(s)
```

Please

1. Run this program in the stacker by clicking one of the [Run ▶] buttons above;
2. The stacker would show how this program produces its result(s);
3. Keep clicking ⏭ Next until you reach a configuration that you find particularly helpful;
4. Click 🔗Share This Configuration to get a link to your configuration;
5. Submit your link below;

https://smol-tutor.xyz/stacker/?syntax=Lispy&randomSeed=smol-tutor&hole=%E2%80%A2&nNext=2&program=%0A%28deffun+%28addy+x%29%0A++%28%2B+x+y%29%29%0A%28defvar+s+%28addy+1%29%29%0A%28defvar+y+2%29%0A%0As%0A&readOnlyMode= 30

(Task 10 of 10) Please write a couple of sentences to explain how your configuration explains the result(s) of the program. 31

Shows error when addy evaluates y before it is defined 32

Let's review what we have learned in this tutorial. 33

- Variables are bound to values. Specifically, every variable definition evaluates the expression immediately and binds the variable to the value, even if the variable is not used later in the program; every function call evaluates the actual parameters immediately and binds the values to formal parameters, even if the formal parameter is not used in the function.

- Every block evaluates its definitions and expressions in reading order (i.e., top-to-bottom and left-to-right).

You have finished this tutorial 🎉🎉🎉

Please print the finished tutorial to a PDF file so you can review the content in the future. **Your instructor (if any) might require you to submit the PDF.**

Start time: 1780300574795